MEMORANDUM

TO: Marketing Leadership & Executive Team

FROM: Vyshnav P S, Data Analyst

DATE: September 25, 2026

SUBJECT: Strategic Reallocation of Ad Spend: Facebook vs. AdWords

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1. Executive Summary & Business Problem

Our marketing organization faces a critical resource allocation challenge: determining how to distribute our advertising budget between Facebook and Google AdWords to maximize user conversions. Historically, spend has been allocated without a rigorous mathematical framework to prove which platform definitively drives more value, nor how market volatility impacts our overall cost of acquisition.

This programmatic analysis was commissioned to mathematically identify the superior platform, test our ability to forecast future performance using machine learning, and provide actionable, data-driven recommendations for budget optimization over the next fiscal year.

2. Analytical Methodology

An end-to-end data pipeline was engineered to ingest and analyze roughly three years (658 distinct campaign days) of historical performance data. After rigorous data quality assurance—verifying zero missing or duplicated records—the analysis utilized three distinct statistical methodologies:

- **Welch's Two-Sample T-Test:** To determine if the variance in conversions between platforms was mathematically significant rather than a random artifact.
- **Linear Regression Modeling:** To test if we can accurately predict future conversion volumes based strictly on the number of clicks purchased.
- **Time-Series Cointegration:** To evaluate if macroeconomic or seasonal factors affect both platforms simultaneously or independently.

3. Baseline Performance Metrics

Facebook Ads	Google AdWords
11.97	**5.93**
Average Daily Conversions at $4.31 Avg CPC	Average Daily Conversions at $2.27 Avg CPC

The A/B Hypothesis testing confirms absolute statistical significance (*p-value*: 6.88e-237). The data

4. Advanced Analytics & Machine Learning Insights

Following the baseline A/B test, a Predictive Linear Regression model was trained to forecast Facebook conversions based on click volume. This model is critical for determining how we forecast future campaign ROI.

> **Critical Finding: Clicks are a "Vanity Metric"**
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> The machine learning model returned an **R-squared value of -0.0365**. This reveals a complete lack of linear relationship between Facebook ad clicks and actual conversions. Purchasing a higher volume of clicks does not reliably guarantee a proportional increase in conversions.

Finally, a Time-Series Cointegration analysis was conducted to assess platform risk over time *(p-value: 0.0000)*. The mathematical results confirm that both Facebook and AdWords are tethered to the same underlying macroeconomic forces. They are structurally cointegrated; when market conditions cause conversions to dip on Facebook, AdWords will experience the exact same downturn simultaneously.

5. Strategic Recommendations

Based on the mathematical evidence gathered across the 1,000-day operational dataset, I advise the following immediate adjustments to our marketing strategy:

- **Reallocate Primary Budget to Facebook Ads:** Shift the majority of ad spend toward Facebook to capture the proven 2x conversion volume multiplier. However, marketing teams must implement strict, automated daily CPC caps to control the platform's high cost volatility.
- **Abandon Click-Based Forecasting KPIs:** Immediately cease using "Cost Per Click" (CPC) or total click volume as key performance indicators for predicting revenue or campaign success. Transition the analytics and media-buying teams to track Cost Per Acquisition (CPA) exclusively, as our regression model proves clicks do not equal sales.
- **Develop Alternative Marketing Hedges:** Because AdWords and Facebook share the exact same market volatility, AdWords cannot be used as a "safe hedge" when Facebook underperforms. The organization must explore non-cointegrated marketing channels—such as SEO, direct email marketing, or physical mailers—to diversify risk during seasonal downturns.